



Rahul Choudhary
Podcaster | Tech Content Creator | Developer | Influencer

Node.js

ROADMAP



Your Guide to Becoming a Node.js Developer

1. JavaScript Fundamentals (Must Before Node.js)

Node.js runs on JavaScript, so strong JS basics are compulsory.

Learn:

Variables (**var, let, const**)

Data Types (**string, number, boolean, null, undefined**)

Operators & Conditions

Loops (**for, while, forEach**)

Functions (**normal, arrow**)

Arrays & Objects

Scope & Hoisting

Closures

Promises

Async / Await

ES6+ features (**spread, rest, destructuring, modules**)



Goal: Write clean, modern JavaScript code.



2. Introduction to Node.js

Get comfortable with Node.js environment.

Learn:

What is Node.js

Node.js vs Browser JavaScript

V8 Engine

Installing Node.js & npm

Running first Node.js app

REPL

Global objects (**`__dirname`**, **`__filename`**)

 **Goal: Run JS outside the browser.**



3. Node.js Core Modules

Core modules are the foundation of Node.js.

Learn:

fs (File System)

path

os

http

events

url

crypto



Goal: Handle files, servers, paths, and system operations.



4.NPM & Package Management

Learn how to use external libraries.

Learn:

npm basics

package.json

Dependencies vs DevDependencies

Semantic Versioning

Common packages (**nodemon**, **dotenv**)

Scripts in npm

 **Goal: Manage third-party libraries efficiently.**



5. Creating Server with Node.js

Understand backend server creation.

Learn:

HTTP server using **http** module

Handling routes manually

Request methods (GET, POST, PUT, DELETE)

Parsing request body

Sending JSON response

 **Goal: Build basic backend logic without frameworks.**



6. Express.js (Most Important)

Express makes Node.js development fast and easy.

Learn:

What is Express

Express setup

Routing

Middleware

Request & Response objects

Error handling

Static files

Express Router

 **Goal: Build scalable APIs quickly.**



7. RESTful API Development

Create professional APIs.

Learn:

REST architecture

CRUD operations

Route naming conventions

Controllers & services

Pagination

Filtering & sorting

API versioning

 **Goal: Build clean and reusable APIs.**



8. Database

(MongoDB + Mongoose)

Backend is incomplete without database.

Learn:

What is Database

SQL vs NoSQL

MongoDB basics

Collections & Documents

CRUD operations

Mongoose schemas & models

Relationships

Indexing

 **Goal: Store and manage data properly.**



9. Authentication & Authorization

Security is critical.

Learn:

Authentication vs Authorization

JWT (JSON Web Token)

Password hashing (bcrypt)

Login & Register APIs

Role-based access

Refresh tokens

 **Goal: Secure your application.**



10. Middleware & Advanced Express

Improve backend architecture.

Learn:

Custom middleware

Global error handler

Request validation

Rate limiting

CORS

Helmet

 **Goal: Write production-ready code.**



11. File Upload & Cloud Storage

Handle real-world features.

Learn:

Multer

File validation

Image upload

Cloudinary / AWS S3 basics

File security

 **Goal: Manage user files efficiently.**



12. Environment & Configuration

Manage different environments.

Learn:

.env files

Environment variables

Dev vs Production setup

Config files structure

 **Goal: Safe and flexible deployment.**



13. Testing & Debugging

Ensure application quality.

Learn:

Unit testing

Integration testing

Jest / Mocha

API testing with Postman

Debugging Node.js

 **Goal: Build bug-free applications.**



14. Performance & Optimization

Make your app faster.

Learn:

Event Loop

Async operations

Clustering

Caching (Redis basics)

Memory leaks

 **Goal: High-performance backend.**



15. Deployment & DevOps Basics

Push your app live.

Learn:

Hosting options (Render, Railway, AWS)

Build process

PM2

Reverse proxy (Nginx basics)

CI/CD basics

 **Goal: Deploy real projects.**

